SIT720 6.2C CREDIT TASK

Classification Models - Gallstone Disease Classification

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Unit	SIT720 Machine Learning
Dataset	UCI Gallstone (dataset ID 1150)
Models	Logistic Regression and Random Forest

Executive summary. The executed dataset contains 319 records and 38 predictors with no missing values. The classes are almost perfectly balanced (159 gallstone-present and 160 gallstone-absent). Under the required 70:30 stratified split, tuned Logistic Regression achieved the strongest test ROC-AUC (0.814), with accuracy 0.729, F1 0.723 and disease recall 0.708. Random Forest achieved ROC-AUC 0.736 and showed a larger cross-validation train-validation gap. When the split was changed to 80:20 and both models were re-tuned, Logistic Regression still had the higher ROC-AUC (0.816 vs 0.773), although Random Forest had better accuracy and F1 at the default threshold on that split. The final recommendation is Logistic Regression as the stronger model for this experiment, with clear limitations for any clinical deployment.

All numerical results, figures and tables in this report come from the executed Jupyter notebook supplied with the submission.

1. Understanding the Classification Problem

Problem definition. The task is binary classification: predict whether gallstones are present from non-imaging patient information. The official UCI variable documentation codes *Gallstone Status* as 0 = present and 1 = absent. For clearer evaluation, the notebook recodes the positive class as *Gallstone_Present = 1*. This means precision, recall and F1 refer to identifying disease-positive patients.

Inputs. There are 38 predictors covering demographics (for example age and gender), comorbidity indicators, body-composition/bioimpedance measurements (BMI, body water, fat and muscle measures), and laboratory measurements (glucose, lipids, liver enzymes, creatinine, GFR, CRP, haemoglobin and vitamin D).

Output. One binary target: gallstone disease present or absent. Binary classification allows use of disease-class precision, recall and F1, while ROC-AUC evaluates ranking performance across all probability thresholds.

Practical significance and errors. A non-imaging risk model could support screening or prioritisation for further clinical assessment, but it cannot replace imaging or medical judgement. A false negative is important because a person with gallstones is labelled as disease-absent, which could reduce the urgency of follow-up. A false positive may lead to unnecessary concern or additional testing. Therefore recall is considered alongside ROC-AUC, accuracy and F1.

Assumption. Records are treated as independent patients, and the local CSV is treated as the analysis dataset. The model is evaluated as a predictive experiment, not as a validated clinical diagnostic tool.

2. Data Exploration and Preparation

Item	Executed result
Rows	319
Predictors	38
Target columns	1
Missing entries	0
Duplicate rows	0
Gallstone present	159 (49.84%)
Gallstone absent	160 (50.16%)
70:30 training set	223 records
70:30 test set	96 records

Feature types. Seven variables are treated as categorical/binary/ordinal-coded: Gender, Comorbidity, Coronary Artery Disease, Hypothyroidism, Hyperlipidemia, Diabetes Mellitus and Hepatic Fat Accumulation. The remaining 31 variables are treated as continuous/integer measurements. Age is numeric and is modelled as continuous.

Missing values and preprocessing. No missing values were observed. Median imputation for numeric variables and most-frequent imputation for categorical variables are kept inside the pipeline as defensive steps; on this dataset they do not change any values. Median is robust to extreme numeric values. Numeric variables are standardised for Logistic Regression, and categorical/ordinal-coded variables are one-hot encoded. All transformations are fitted only on training folds, preventing data leakage.

Split strategy. A stratified 70:30 split with random state 42 is used so the class proportion remains stable in training and test data. The 30% test set is untouched during tuning. Hyperparameters are chosen by stratified five-fold cross-validation on the 70% training set.

Relevant preprocessing code

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, stratify=y, random_state=42)

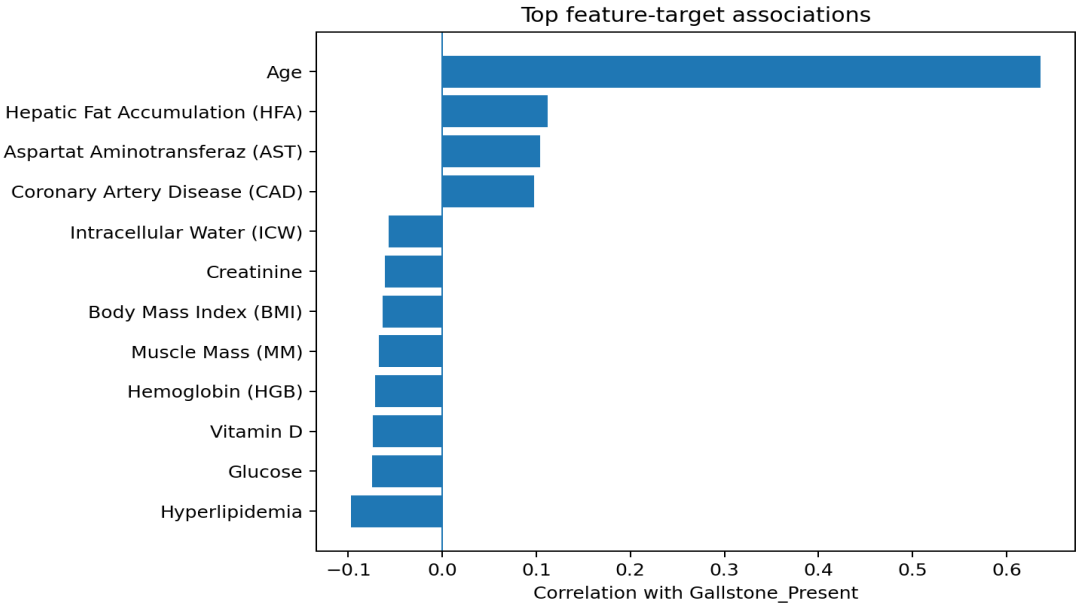
preprocessor = ColumnTransformer([
    ('num', Pipeline([('imputer', SimpleImputer(strategy='median')),
                      ('scaler', StandardScaler())]), continuous_features),
    ('cat', Pipeline([('imputer', SimpleImputer(strategy='most_frequent')),
                      ('encoder', OneHotEncoder(handle_unknown='ignore'))]),
     categorical_features)
])
```

Preprocessing limitations. One-hot encoding Comorbidity and HFA does not use their natural ordering, so an ordinal encoding could be explored in future work. Standardisation is unnecessary for Random Forest but is harmless inside a shared comparison pipeline. The sample is also small and comes from one clinical setting, limiting generalisability.

2.1 Exploratory Analysis

The strongest marginal relationship with the recoded disease target is Age (correlation 0.636). The remaining marginal correlations are much smaller: HFA 0.112, AST 0.104, CAD 0.098 and Hyperlipidemia -0.097. No pair of continuous predictors had absolute correlation at or above 0.85, so no feature was removed solely for severe collinearity. Correlation is used here as an exploratory indicator, not proof of causality or independent predictive importance.

Feature	Correlation with disease
Age	0.636
Hepatic Fat Accumulation (HFA)	0.112
Aspartat Aminotransferaz (AST)	0.104
Coronary Artery Disease (CAD)	0.098
Hyperlipidemia	-0.097
Glucose	-0.075
Vitamin D	-0.074
Hemoglobin (HGB)	-0.072



Informative and potentially redundant features. Age is clearly the most informative single feature in the marginal analysis. Model-based evidence is consistent: the tuned Logistic Regression gives Age the largest absolute coefficient (1.505 after scaling), while Random Forest gives Age the largest impurity importance (0.291). Because the remaining predictors have weaker marginal associations, nonlinear interactions may still contribute even when simple correlations are small.

3. Model Development and Selection

Logistic Regression. This model was selected as an interpretable linear probabilistic baseline. It assumes that predictors have an approximately linear relationship with the log-odds after preprocessing and benefits from regularisation when many features are present. Strengths are simplicity, fast prediction and interpretable coefficients. Weaknesses are limited ability to represent nonlinear interactions and sensitivity to misspecification.

Random Forest. This model was selected as a nonlinear tree ensemble. It makes no linearity assumption and can capture interactions automatically. Its strengths are flexibility and robustness to monotonic transformations. Its weaknesses are lower transparency, greater model complexity and the possibility of overfitting a small dataset.

Validation metric. ROC-AUC is the primary tuning metric because it measures probability ranking across thresholds. Accuracy, balanced accuracy, precision, recall and F1 are reported to describe threshold-based behaviour. Balanced accuracy is included even though the dataset is balanced, so the evaluation remains appropriate if class proportions change.

Model	Hyperparameters explored	Reason
Logistic Regression	C = 0.1, 1, 10; penalty = L1 or L2	Control regularisation strength and sparsity.
Random Forest	trees = 100/200; depth = None/6; min leaf = 1/3; max features = sqrt	Control ensemble stability and tree complexity.

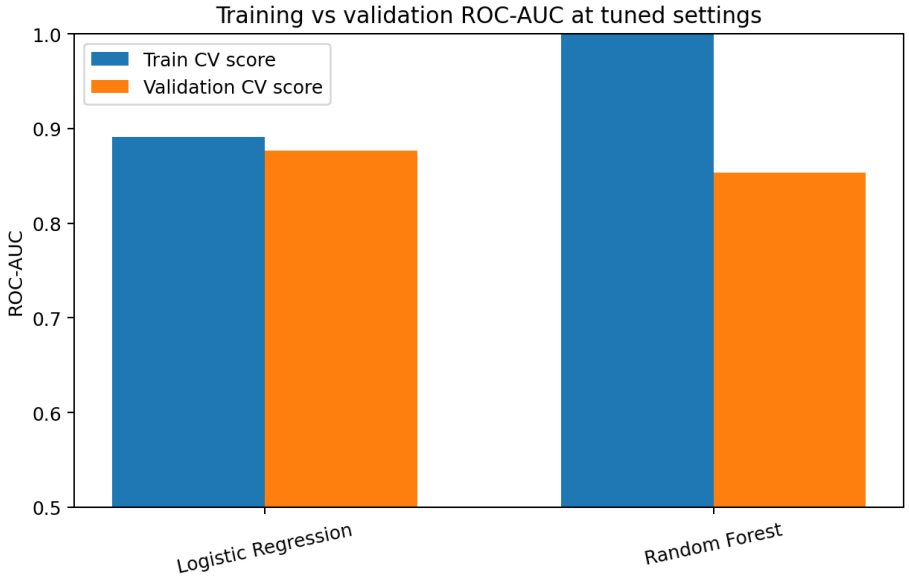
Relevant tuning code

```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
search = GridSearchCV(model_pipeline, parameter_grid,
                      cv=cv, scoring='roc_auc', refit=True)
search.fit(X_train, y_train)
```

Cross-validation diagnosis

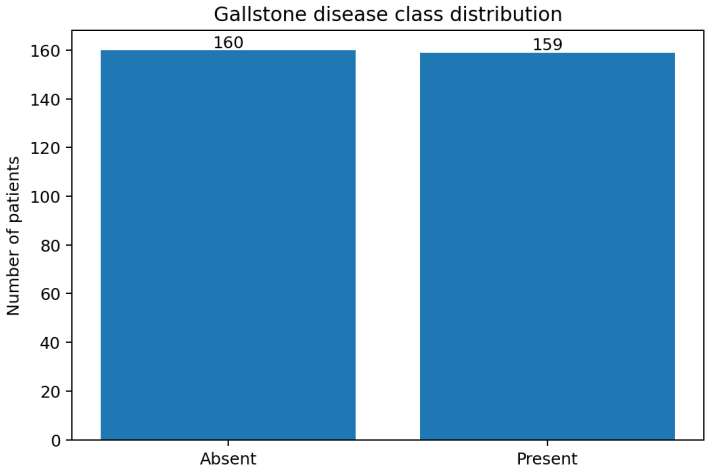
Model	Train AUC	CV AUC	Gap	Best setting	Diagnosis
Logistic Regression	0.891	0.877	0.014	C=0.1, L1	Generalises reasonably
Random Forest	1.000	0.853	0.147	200 trees, depth=None, min leaf=3	Possible overfitting

Logistic Regression has only a 0.014 train-validation AUC gap (0.891 vs 0.877), indicating stable generalisation within cross-validation. Random Forest reaches training AUC 1.000 but validation AUC 0.853, leaving a much larger 0.147 gap. This is quantitative evidence of excess model complexity and possible overfitting.



4. Addressing Data Imbalance

The class counts are 160 absent and 159 present, giving a majority/minority ratio of 1.006 and a minority share of 49.84%. This is effectively balanced. Therefore no SMOTE, undersampling or class weighting was applied. Artificial balancing would add complexity without correcting a real class-frequency problem and could distort a small clinical dataset.



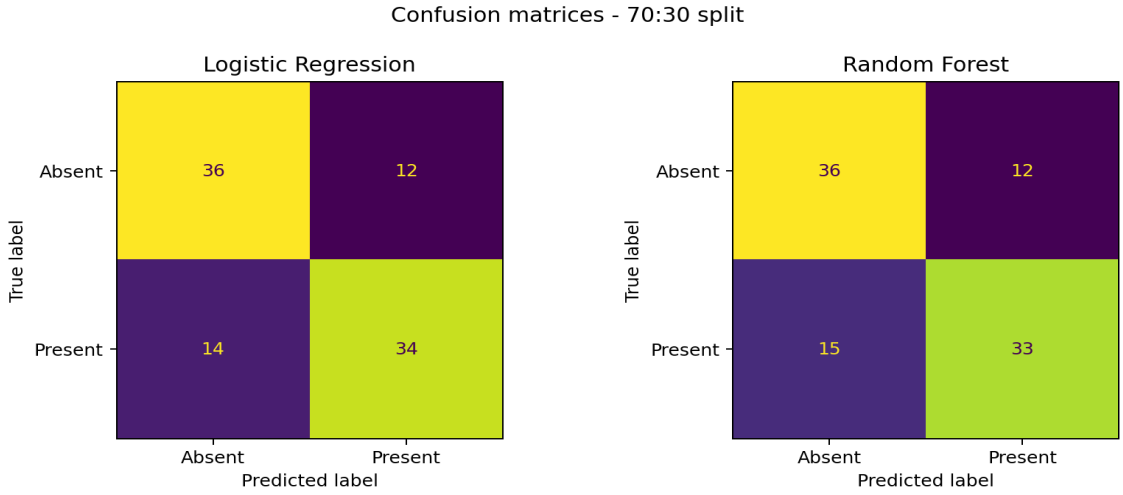
Because no material imbalance exists, the "before versus after balancing" comparison is not applicable. The decision is supported directly by the observed class counts.

5. Final Model Evaluation and Recommendation

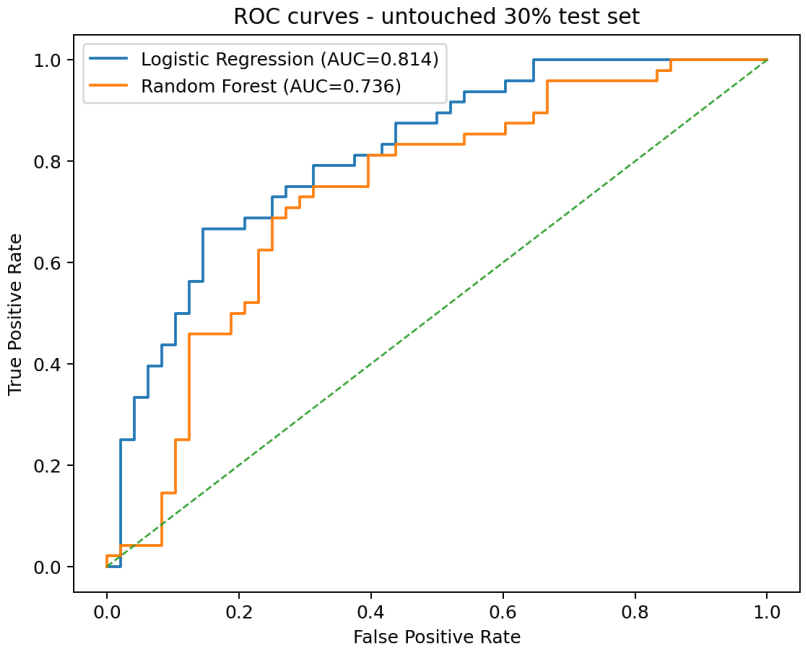
Untouched 30% test set. After selecting hyperparameters by cross-validation on the training set, both fitted models were evaluated once on the 96-record test set.

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.729	0.729	0.739	0.708	0.723	0.814
Random Forest	0.719	0.719	0.733	0.688	0.710	0.736

Logistic Regression is the better model under the primary criterion: ROC-AUC 0.814 versus 0.736 for Random Forest. It also has slightly higher accuracy (0.729 vs 0.719), F1 (0.723 vs 0.710) and disease recall (0.708 vs 0.688).



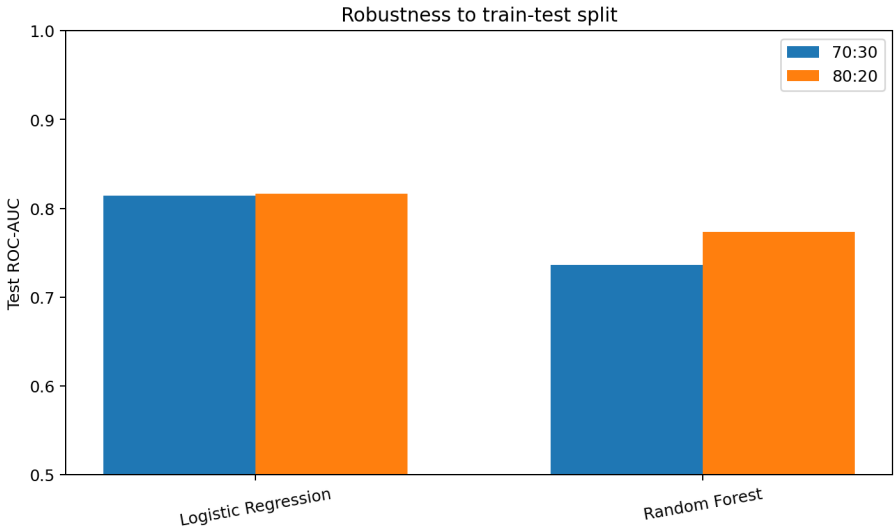
Error analysis. Logistic Regression produced 36 true negatives, 12 false positives, 14 false negatives and 34 true positives. Random Forest produced 36 true negatives, 12 false positives, 15 false negatives and 33 true positives. The selected model therefore missed one fewer disease-positive patient. Since false negatives can be more consequential in a screening context, this supports the Logistic Regression choice, although the decision threshold would need clinical calibration.



5.1 Robustness: 80:20 Re-evaluation

The experiment was repeated with a stratified 80:20 split, using the same five-fold cross-validation and re-tuning process. This changes the held-out patients and increases the training sample, providing a simple robustness check.

Model	70:30 AUC	80:20 AUC	70:30 Accuracy	80:20 Accuracy	70:30 F1	80:20 F1
Logistic Regression	0.814	0.816	0.729	0.688	0.723	0.667
Random Forest	0.736	0.773	0.719	0.734	0.710	0.721



Using the pre-declared primary metric, Logistic Regression remains better: ROC-AUC 0.816 versus 0.773. Therefore the main ranking is preserved. However, Random Forest has higher 80:20 accuracy (0.734 vs 0.688) and F1 (0.721 vs 0.667) at the default 0.5 threshold. This difference shows that model ranking is metric- and threshold-sensitive. With only 319 records, a change in held-out patients can noticeably change threshold-based results even when overall ranking quality is stable.

5.2 Computational Cost, Robustness and Deployment

Measured 70:30 grid-search times were approximately 5.6 seconds for Logistic Regression and 4.3 seconds for Random Forest in this execution. On the 80:20 run they were approximately 0.3 seconds and 4.2 seconds respectively. These small wall-clock values are affected by process and parallelisation overhead, so they should not be treated as a precise complexity benchmark. Structurally, Logistic Regression is simpler to store, interpret and score, while Random

Forest must evaluate many trees.

Final recommendation. Logistic Regression is recommended for this experiment because it has the highest 70:30 test ROC-AUC, better disease recall and F1, a smaller cross-validation generalisation gap, and the highest ROC-AUC again under 80:20. It also provides clearer coefficients. Random Forest remains useful as a nonlinear comparison but shows stronger overfitting evidence.

Deployment caution. The numerically better model is not automatically suitable for clinical deployment. Before use it would require external validation on a different population, probability calibration, threshold selection based on clinical costs, fairness and subgroup analysis, privacy review, drift monitoring, and prospective evaluation. The model should support rather than replace clinical diagnosis.

Relevant final-evaluation code

```
pred = best_model.predict(X_test)
prob = best_model.predict_proba(X_test)[:, 1]
auc = roc_auc_score(y_test, prob)
f1  = f1_score(y_test, pred)
recall = recall_score(y_test, pred)
```

6. Reflection on Machine Learning Model Development

The main challenge was making a fair comparison on a small clinical dataset without allowing test-set information to influence model selection. A stratified split preserved the class ratio, while five-fold cross-validation was used only on the training data for hyperparameter selection. A second challenge was handling mixed feature types without leakage. Continuous variables were imputed and standardised inside the pipeline, while categorical and ordinal-coded variables were imputed and one-hot encoded.

The results demonstrated the trade-off between predictive performance and complexity. Logistic Regression was easier to interpret and showed a small train-validation ROC-AUC gap, whereas Random Forest achieved perfect training ROC-AUC but a lower validation score, indicating overfitting risk. The 80:20 experiment showed that Logistic Regression retained the best ROC-AUC, although Random Forest had better threshold-based accuracy and F1 on that particular test split. This illustrates why conclusions should not depend on one metric or one partition.

Future work should use larger external cohorts, nested cross-validation, probability calibration, clinically justified decision thresholds, fairness analysis, and prospective validation before any clinical deployment.

Reflection word count: 170 words.

GenAI Acknowledgement

ChatGPT was used to assist with planning the analysis, creating an initial code scaffold, debugging, checking coverage of the assessment brief, and editing the written explanations. The notebook was executed on the Gallstone dataset and the reported metrics and figures are generated by the code. The student is responsible for critically reviewing the analysis, understanding the modelling decisions, and ensuring the final submitted work complies with Deakin academic-integrity requirements.

References

[1] UCI Machine Learning Repository, *Gallstone*, dataset ID 1150. Variable documentation identifies 38 predictors, no missing values, and target coding 0 = gallstones present and 1 = absent.

[2] I. Esen, H. Arslan, S. Aktürk Esen, M. Gül■en, N. Kültekin and O. Özdemir, “Early prediction of gallstone disease with a machine learning-based method from bioimpedance and laboratory data,” *Medicine*, 2024, doi: 10.1097/MD.0000000000037258.

[3] Deakin University, SIT720 Task 6.2C assessment brief: Classification Models, supplied with this task.

Submission note: The accompanying Jupyter notebook contains the complete runnable code, all outputs and the saved figures used in this report.